School of Computing and Engineering
National Institute of Business Management
Kandy

Higher National Diploma in Software Engineering

HNDSE 25.1F

AgroSmart: -

An IoT-Based Smart Precision Agriculture System with
Cloud Monitoring and Multi-Sensor Control

Internet of Things – Proposal

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Prepared for –

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Date of Submission – 11/01/2026

Declaration

We, the undersigned, hereby declare that this proposal report for AgroSmart: IoT-Based Smart Precision Agriculture System with Cloud Monitoring and Multi-Sensor Control has been prepared by us (**Group 05**) as part of a group coursework for the Internet of Things module at the National Institute of Business Management, Kandy.

We confirm that:

1. The work presented is our own and has not been submitted previously for any other academic award.
2. All sources of information, ideas, and data from other authors have been properly cited and referenced.
3. The project is carried out in accordance with the institution's academic integrity policies.

We take full responsibility for the content and authenticity of this report.

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Abstract

This project, AgroSmart: IoT-Based Smart Precision Agriculture **System with Cloud Monitoring and Multi-Sensor Control**, aims to develop a **smart agriculture monitoring system** integrated with a **precision agriculture module** to assist farmers in making data-driven decisions. The system collects real-time environmental data, including temperature, humidity, soil moisture, light intensity, and water levels, using IoT sensors connected to an **ESP32 microcontroller**. Data is uploaded to a **cloud platform (Firebase)** and visualized through a **web and mobile dashboard**.

The Smart Precision Agriculture module analyzes collected data using **AI and machine learning algorithms** to provide actionable insights, such as optimal irrigation schedules, crop selection recommendations, and predictive trends for seasonal climate variations. The project is initially focused on the Kandy region in Sri Lanka, but is designed to be modular and expandable for additional sensors and features in the future.

The outcomes of this project demonstrate the potential of combining **IoT, cloud computing, and AI** for efficient resource management and sustainable agricultural practices. This project is carried out as a **group coursework** and provides both practical implementation and academic insights into smart farming technologies.

Acknowledgement

We want to express our sincere gratitude to **Mr. H. M. H. T. Hikkaduwa**, our lecturer, for their continuous guidance, encouragement, and valuable suggestions throughout the preparation of this proposal report.

We also wish to thank the **NIBM staff** for providing access to resources, facilities, and support necessary for developing the project concept.

Finally, we extend our gratitude to all **team members** for their dedication, collaboration, and commitment in completing this group coursework. This project would not have been possible without the cooperation and efforts of everyone involved.

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Executive Summary

AgroSmart is an **IoT-based Smart Agriculture Monitoring System integrated with a Smart Precision Agriculture Module** designed to assist farmers by providing **real-time environmental monitoring and AI-based actionable insights**. The system will monitor **temperature, humidity, soil moisture, light intensity, and water levels** using a network of sensors connected to an **ESP32 microcontroller**. Data will be uploaded to the **Firebase cloud** and displayed on a **web and mobile dashboard**.

The project aims to **enhance irrigation efficiency, optimize crop selection, and forecast seasonal trends through the use of AI/ML analysis on** collected sensor data. AgroSmart will serve as a modular platform, allowing additional features and sensors to be integrated in the future.

Expected outcomes include:

- Real-time environmental data collection.
- Cloud-based visualization dashboard and mobile application.
- AI/ML-assisted recommendations for irrigation and crop selection.
- Academic documentation, including reports and presentations.

Project Background

Agriculture in regions like **Kandy, Sri Lanka**, faces multiple challenges, including inefficient water usage, unpredictable weather patterns, and reduced crop yield. Traditional farming methods **lack real-time monitoring and predictive insights**, resulting in water wastage, crop loss, and lower productivity.

With the advancement of **IoT, cloud computing, and AI**, it is now possible to monitor environmental factors in real-time, analyze data, and provide actionable recommendations. Implementing a **Smart Agriculture Monitoring System** can help farmers make **data-driven decisions**, optimize water usage, improve crop yields, and reduce resource waste.

The **motivation** behind AgroSmart is to develop an **academic-level prototype** that demonstrates the potential of **IoT and AI in precision agriculture** while providing a foundation for future enhancements.

Project Aim and Objectives

Project Aim

To develop a **Smart Agriculture Monitoring System** integrated with a **Smart Precision Agriculture module** that collects real-time environmental data, analyzes it using AI/ML, and provides actionable insights for farmers.

Project Objectives

1. Design and implement an **IoT sensor network,** including:
   - Temperature & Humidity Sensor (DHT11).
   - Soil Moisture Sensors (Capacitive & Normal).
   - Light Intensity Sensor (LDR/MH-LDR).
   - Ultrasonic Sensor for water level detection.
   - Relay-controlled pump for smart irrigation.
2. Collect real-time data and store it on the **Firebase cloud platform.**
3. Develop a **dashboard and mobile application** for the visualization of sensor data.
4. Integrate **AI/ML algorithms** to provide:
   - Irrigation recommendations based on humidity and soil moisture.
   - Crop selection suggestions.
   - Seasonal climate trends and predictive insights.
5. Automate irrigation based on analyzed sensor data.
6. Ensure **modular system design** to allow future integration of additional sensors or features.

Scope of the Project

1. Target a specific area: **Kandy, Sri Lanka**, for initial implementation.
2. Monitor environmental parameters: **temperature, humidity, soil moisture, light intensity,** and **water levels.**
3. Provide actionable insights for farmers, including:
   - Optimal irrigation schedule.
   - Crop selection recommendations.
   - Predictions of climate and soil trends.
4. Modular and expandable system for additional sensors or AI modules.
5. Academic scope: real-time IoT system and AI-based recommendation system.
6. Focused on **proof-of-concept**, not large-scale industrial deployment.

Methodology

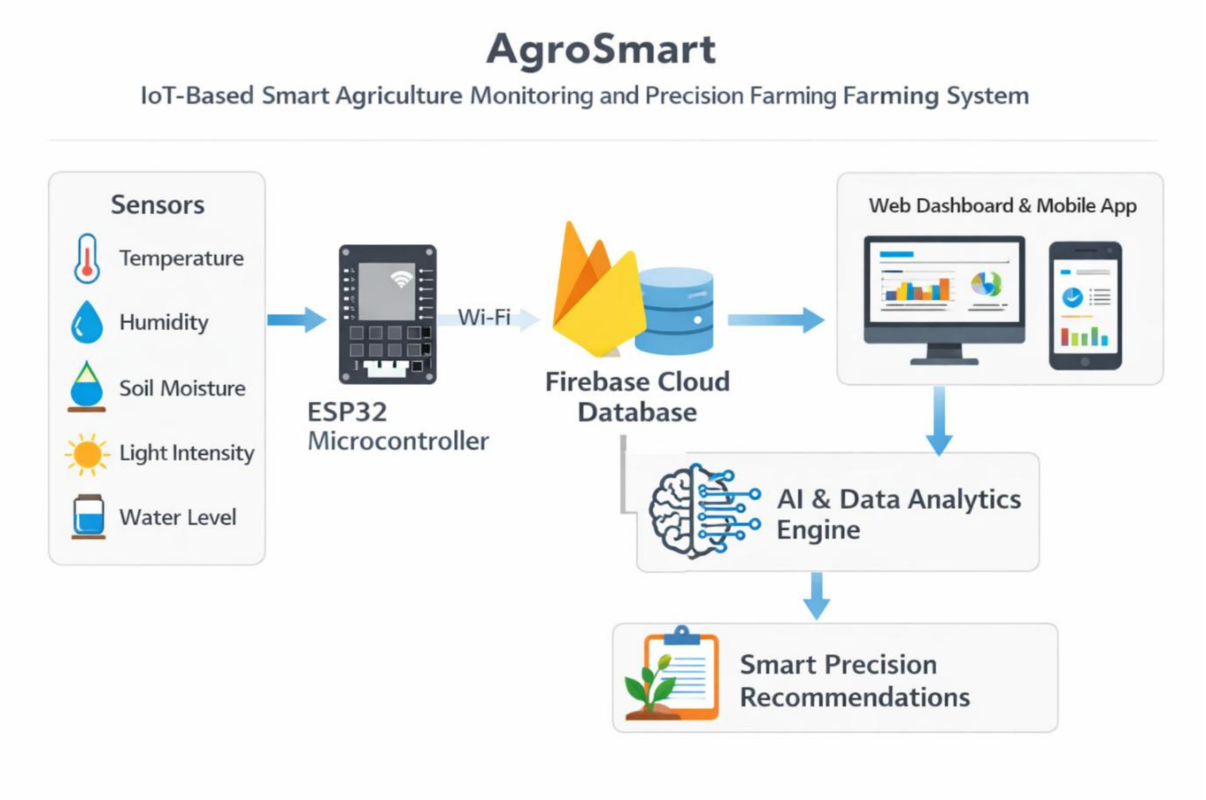


Figure 1: Overall Architecture of AgroSmart System

System Phases

1. **Hardware Layer:**

    - ESP32 microcontroller as the main controller
    - Sensors: DHT11, soil moisture, LDR, ultrasonic
    - Relay module and water pump for automated irrigation
    - Proper power management: 3.3V for low-voltage sensors, 5V external supply for pumps, and high-voltage sensors

2. **Data Collection:**

- Sensors collect real-time environmental data
- ESP32 uploads data to **Firebase Realtime Database**
- A dataset for the historical data

Sensor Name	Parameter Measured	Unit	Location	Purpose
DHT11	Temperature, Humidity	°C, %	Field environment	To monitor ambient environmental conditions affecting crop growth
Soil Moisture Sensor	Soil Moisture Level	Percentage (%)	Soil near plant root zone	To determine soil water content and enable automated irrigation decisions
Ultrasonic Water Level Sensor	Water Level	Centimeters (cm)	Water tank / reservoir	To monitor available water level for irrigation management
LDR Light Sensor	Light Intensity	Lux	Field / crop canopy area	To assess sunlight availability for photosynthesis analysis

Figure 2: Sensor Data Collected by AgroSmart

3. **Data Analysis & AI/ML Module:**

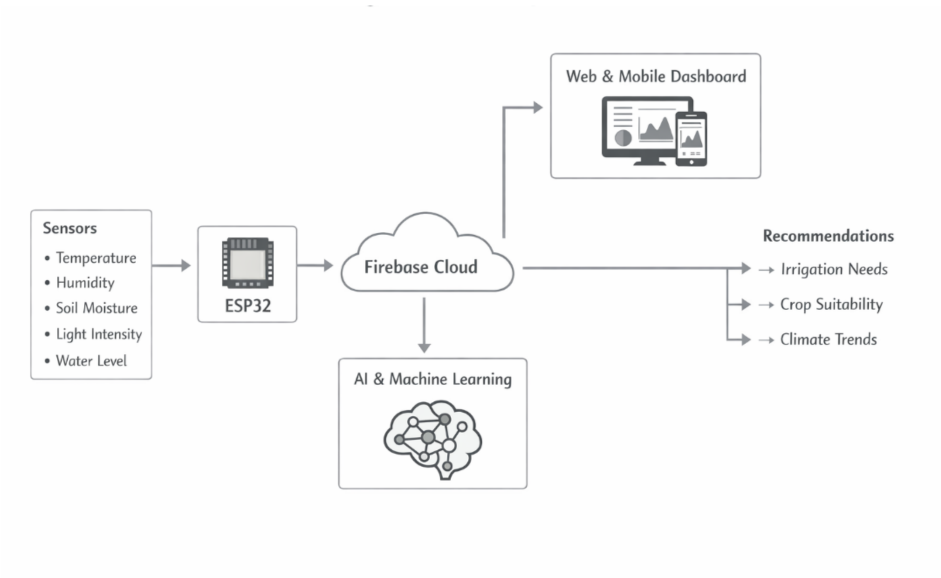


Figure 3: Data Flow from Sensors to Cloud and AI Engine

- Analyze environmental data for irrigation and crop recommendations
- Predict soil moisture trends, humidity patterns, and optimal crop selection
- AI/ML approach:
    - Supervised learning for historically labeled data is available
    - Unsupervised learning for pattern detection in unlabeled data

4. **Dashboard and Mobile Application:**

- Visualize sensor readings in real-time
- Display recommendations and alerts
- Allow remote monitoring of irrigation status

5. **Testing & Validation:**

- Test sensors individually and in combination
- Calibrate soil moisture thresholds and water tank levels
- Validate AI predictions using sample datasets

6. **AI-Based Smart Precision System**

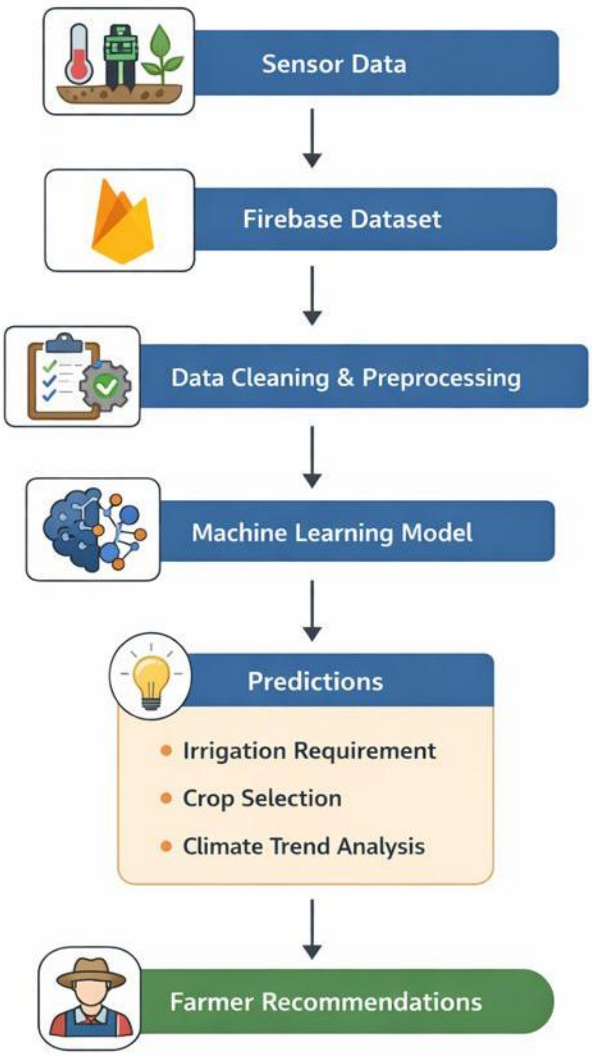


Figure 4: AI and Data Analysis Flow

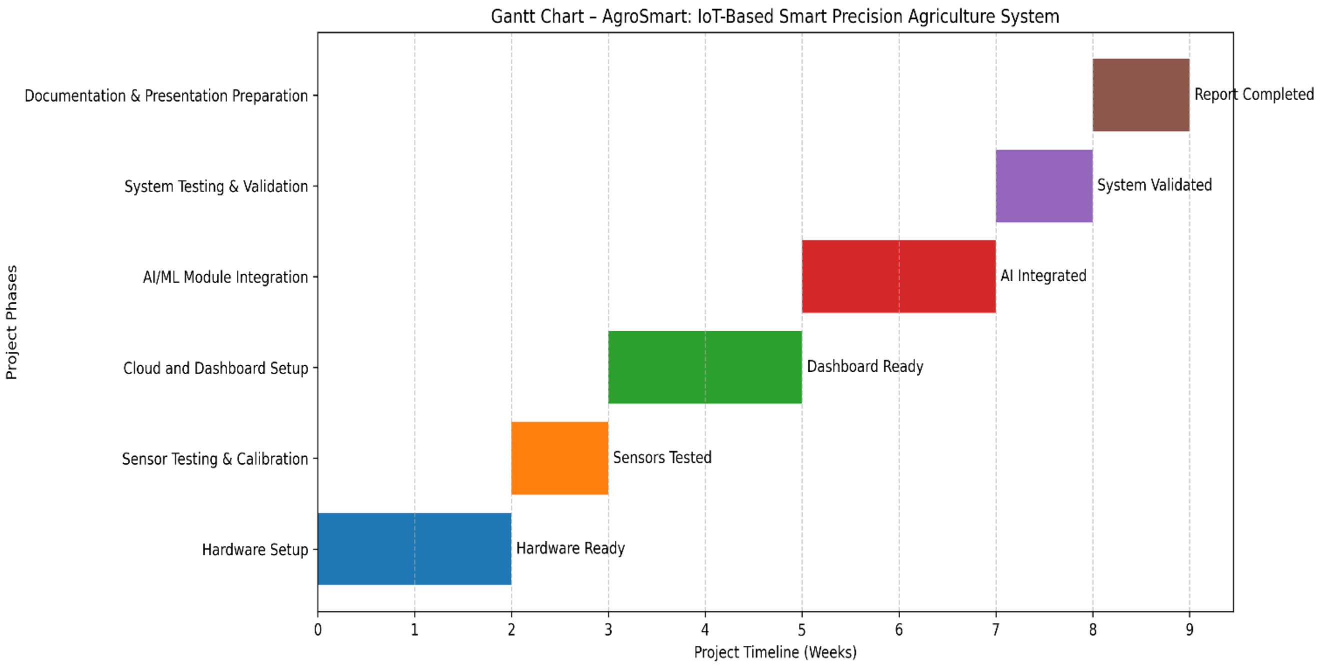
Tools and Technologies

- Microcontroller: ESP32
- Sensors: DHT11, Capacitive/Normal Soil Moisture, LDR, Ultrasonic
- Cloud Platform: Firebase Realtime Database
- AI/ML: Python, TensorFlow / scikit-learn
- Dashboard / App: Web technologies (HTML/CSS/JS) or Flutter for mobile
- Version Control & collaboration: GitHub - Cloud-based Version Control System

Project Deliverables

1. **Hardware Prototype:** ESP32 + sensor network + relay/pump system
2. **Software Module:** Cloud-based data storage, web dashboard, mobile app
3. **AI/ML Module:** Data analytics and recommendations for irrigation and crop selection
4. **Reports & Documentation:** Proposal report, final report
5. **Presentations:** Proposal, progress updates, final presentation

Timeline



Phase	Duration
Hardware setup	2 weeks
Sensor testing & calibration	1 week
Cloud and dashboard setup	2 weeks
AI/ML module integration	2 weeks
System testing & validation	1 week
Documentation & presentation preparation	1 week

References

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Appendices

- **Hardware Block Diagram:** ESP32 connected to sensors, relay, and pump
- **System Architecture Diagram:** Data flow from sensors → ESP32 → Firebase → AI module → Dashboard
- **Sample Sensor Data Table:** Example readings for temperature, humidity, soil moisture, and light intensity
- **Additional Figures/Charts:** Sensor calibration charts, optional images of sensors